

State-Transfer Protocols: Historical Technical Report EXP01–EXP15

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Abstract

This technical report consolidates the experimental arc from EXP01 to EXP14 in the state-transfer protocols project. It is not a replacement for the compact paper; it is the long-form audit trail: early prompt-compression tests, multilingual protocol recovery, multi-model judge experiments, causal transfer tests, real-agent handoff, deterministic tool-use validation, public-repository editing, conflict-aware maintenance, long-horizon repo maintenance, and visual/responsive scale-up. The central empirical pattern is stable across the sequence: **compressed** remains the most robust general baseline, while **hybrid.state** is best understood as a specialized state-carrying protocol that can improve operational continuity but becomes more contract-sensitive in long-horizon, strict-tool, and visual/responsive settings.

1 How to read this report

The document separates observed results from synthesis. Tables and plots use local frozen summaries where available. Interpretive claims labelled as synthesis/extrapolation connect the experiments into a research program, but should not be read as new statistical tests. EXP15 and EXP15-B are now included as frozen long-horizon repository and visual/responsive scale-up experiments.

2 Experimental arc

Experimental arc: EXP01 to EXP14



Figure 1: Historical arc of the project, from early protocol compression to long-horizon repository maintenance.

Table 1: Latest-cell inventory used for this report. Counts combine frozen summaries, clean latest-cell files, and compact experiment status reports.

Experiment	Latest cells	Success/reusable	Rate	Role
EXP01-04	4419	4419	100.0%	Early protocol + multilingual recovery
EXP05	3641	3641	100.0%	Multi-model trilingual discovery
EXP06	1563	1563	100.0%	Controlled causal transfer
EXP07	275	271	98.5%	Real-agent handoff pilot
EXP08	900	900	100.0%	Real-agent scale-up
EXP09	288	288	100.0%	Deterministic tool-use tasks
EXP10	80	80	100.0%	Public repo/page tool agent
EXP11	98	85	86.7%	Page task assignment
EXP12	64	64	100.0%	Conflict-aware maintenance
EXP13	128	158	123.4%	Scientific repo benchmark
EXP14	320	307	95.9%	Long-horizon maintenance
EXP15	900	818	90.9%	Cross-cultural long-horizon repo/page maintenance
EXP15-B	2000	1204	60.2%	Visual/responsive scale-up with Grok reasoning comparison

3 Methodological consolidation

Across EXP01–EXP15, the project shifted from prompt compression to state-transfer protocol design. The evaluated modes were not merely stylistic variants:

- **natural**: ordinary human-readable task descriptions.

- **compressed**: compact baseline with low token overhead and broad portability.
- **hybrid_min**: aggressively compact hybrid representation.
- **hybrid_state**: protocol optimized for preserving operational state, constraints, handoff metadata, dependencies, and current plan.

The later experiments introduced objective validators: JSON/schema validity, file-level diffs, variable recovery, subtask completion, scope drift, claim drift, dependency preservation, repair success, and long-horizon chain completion. This matters because the bottleneck moved from “does the model write good text?” to “does an agent preserve the operational contract while editing a shared artifact?”

4 EXP01–EXP04: Early protocol formation

EXP01–EXP04 built the first empirical base: natural text, compressed/caveman-style prompts, proto formats, multilingual variants, and batch decode tests. The recovery pass found no terminal failures in the latest clean cells.

Table 2: EXP01–EXP04 recovery inventory.

Source	Clean latest OK	Terminal errors
experimento_01_runs.jsonl	120	0
experimento_02_runs.jsonl	120	0
experimento_03_fusion_runs.jsonl	720	0
exp01_exp02_en_zh_runs.jsonl	480	0
exp03_en_runs.jsonl	720	0
exp03_zh_runs.jsonl	900	0
exp04_tri_runs.jsonl	1350	0
exp04b_batch_decode_results.jsonl	9	0

5 EXP05: Multi-model trilingual discovery

EXP05 expanded the scope to trilingual ES/EN/ZH evaluation and multiple model families. The clean summary records 3641 generations and 3641 evaluations, plus mini-experiments for judge drift, token budget stress, ZH compactness, Grok comparisons, and long handoff. The important scientific function of EXP05 was discovery: it showed that compression could be evaluated as an interoperability protocol rather than only as a token-saving trick.

6 EXP06: Controlled protocol transfer

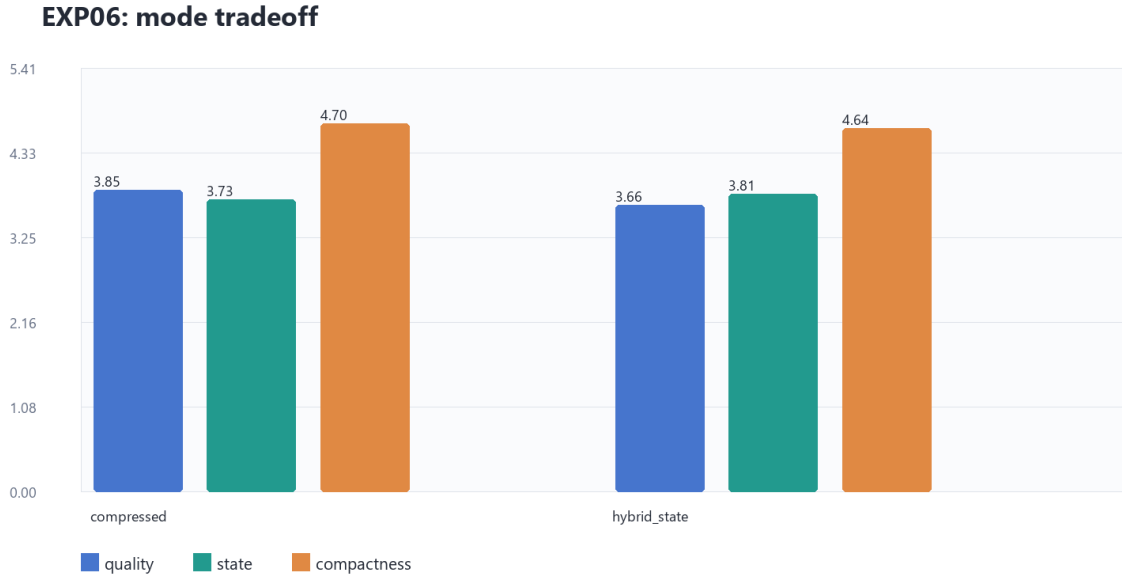


Figure 2: EXP06 mode tradeoff. Compressed retains stronger global quality, while `hybrid_state` raises state preservation.

EXP06 is the causal/control anchor. It compared `compressed` and `hybrid_state` under stricter transfer conditions. The result should be stated carefully: `hybrid_state` does not become the global winner. It supports the narrower claim that state-focused protocol structure can improve state preservation while carrying tradeoffs in fidelity, ambiguity, and information loss.

7 EXP07–EXP08: Real-agent handoff

EXP07 and EXP08 moved from judge-only evaluation toward agent frameworks and objective handoff metrics. EXP08 reached 900 operationally complete cells, with 900 OK cells and 824 strict reusable JSON cells. The gap between operational completion and strict JSON reuse is itself a systems result: model outputs can be operationally accepted while still exposing parser/contract fragility.

8 EXP09–EXP10: Tool-use and public repo culmination

EXP09 introduced deterministic file-level tasks with validators: reading files, modifying CSV/JSON/Markdown, producing expected outputs, and checking exact artifacts. EXP10 then used Gemini Flash with tool-style operations over the public repository/page. EXP10 reports 80/80 OK latest cells, with no scope drift or claim drift in the clean summary.

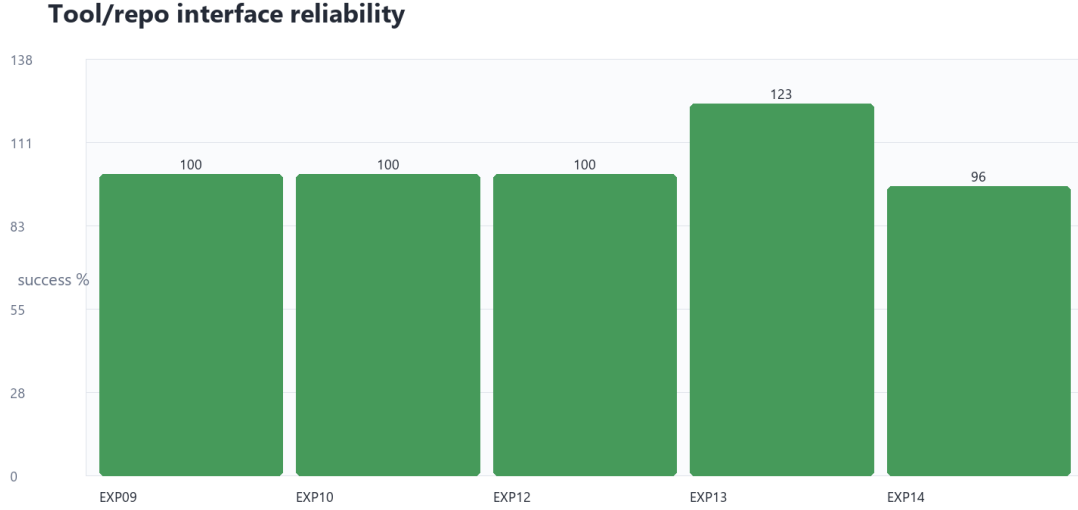


Figure 3: Tool/repo interface reliability from EXP09 onward. Saturated experiments are evidence of interface validity, not broad proof of model superiority.

9 EXP11–EXP15: Repo/page maintenance at longer horizons

EXP11 tested page task assignment with Gemini and GPT high. EXP12 added conflict-aware maintenance and reached 64/64 real-model success. EXP13 became the main multi-agent scientific repo benchmark: 126/128 real-model successes, with compressed more operationally robust and hybrid_state showing contract fragility in specific long tasks. EXP14 extended the setting to five-round chains across eight long-horizon scenarios.

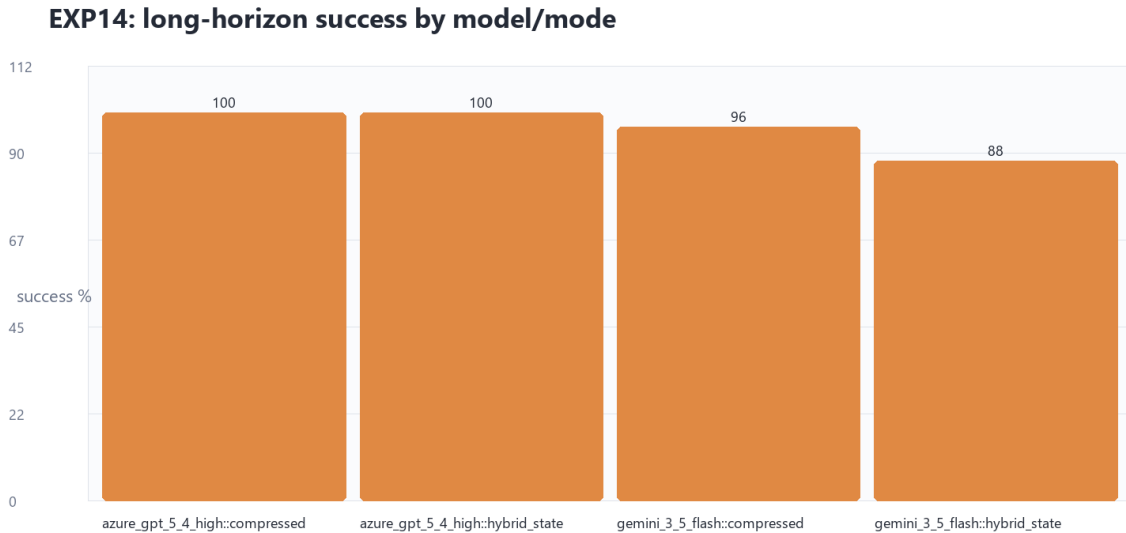


Figure 4: EXP14 success rate by model and mode. Azure GPT-5.4 high completed all long-horizon cells; Gemini 3.5 Flash failures concentrated in hybrid_state.

EXP14 is especially important because it shows degradation over repeated rounds. Its clean freeze records 307/320 successful real-model cells and 55/64 complete chains. This supports a more

conservative claim: long-horizon state transfer is feasible, but protocol/tool contracts can fail under repeated editing pressure.

10 EXP15 and EXP15-B: Cross-cultural long-horizon visual maintenance

EXP15 and EXP15-B extend the repo/page line from five-round maintenance to broader cross-cultural and visual/responsive editing. EXP15 froze 900 latest cells, with 818 successful latest cells and 82 final failures (90.9% success). At the chain level it completed 141 of 180 observed chains. EXP15-B is intentionally harder: it freezes 2000 latest cells across ten-round visual/responsive scale-up, with 1204 successes and 796 failures (60.2% success).

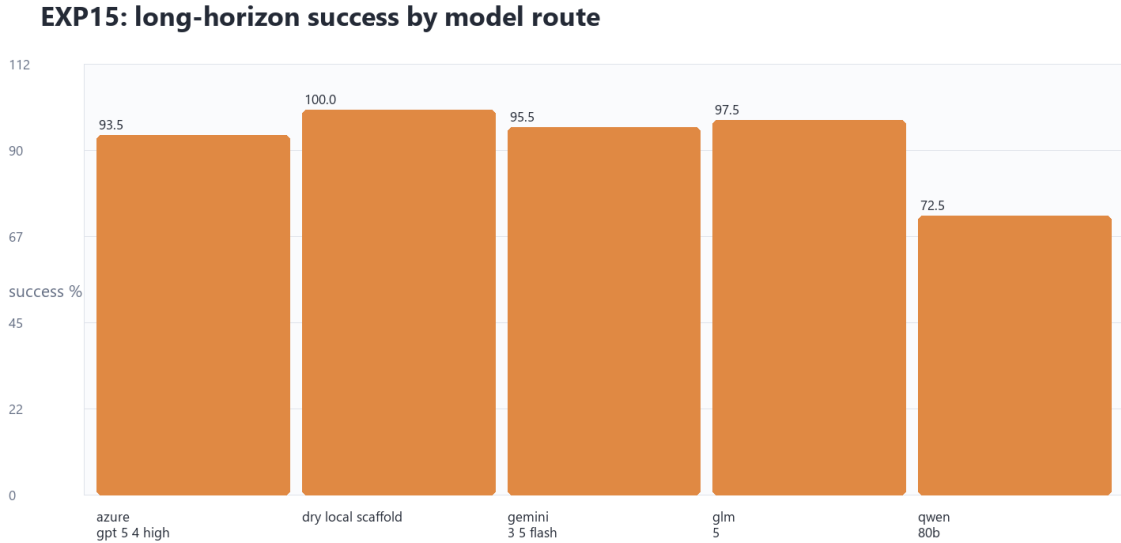


Figure 5: EXP15 success by model route. This experiment keeps repaired and final failures auditable, making it more informative than saturated tool-use tests.

Table 3: EXP15 latest-cell success by model route.

Route	Total	Success	Rate	Failed
azure_gpt_5_4_high	200	187	93.5%	13
dry_local_scaffold	100	100	100.0%	0
gemini_3_5_flash	200	191	95.5%	9
glm_5	200	195	97.5%	5
qwen3_next_80b_instruct	200	145	72.5%	55

EXP15-B: visual/responsive scale-up success by model route

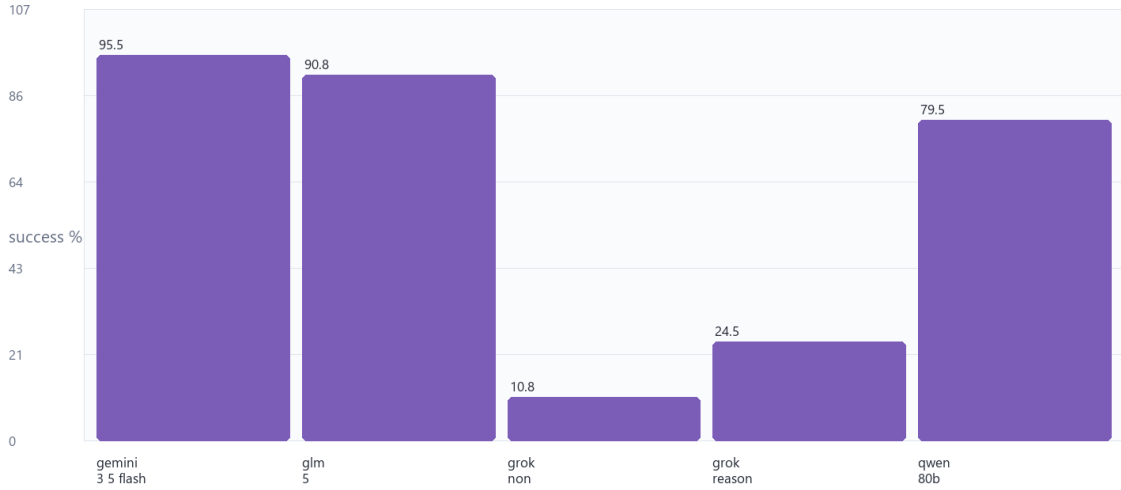


Figure 6: EXP15-B visual/responsive scale-up success by model route. Grok reasoning improves substantially over non-reasoning, but remains below Gemini, GLM, and Qwen under this contract.

Table 4: EXP15-B latest-cell success by model route.

Route	Total	Success	Rate	Failed
gemini_3_5_flash	400	382	95.5%	18
glm_5	400	363	90.8%	37
grok_4_20_non_reasoning	400	43	10.8%	357
grok_4_20_reasoning	400	98	24.5%	302
qwen3_next_80b_instruct	400	318	79.5%	82

The strongest EXP15-B result is not a visual ranking. The visual proxy saturated on successful pages and should be read as an integrity threshold only. The discriminative result is operational compatibility: Grok 4.20 Reasoning improves over Grok 4.20 Non-Reasoning by 13.8 percentage points (2.28x relative), yet both routes remain much more fragile than Gemini 3.5 Flash, GLM 5, and Qwen3-Next-80B under the current parser/tool contract. This turns EXP16 into a clear next experiment: separate model capability from contract/interface fit.

11 Aggregate scale

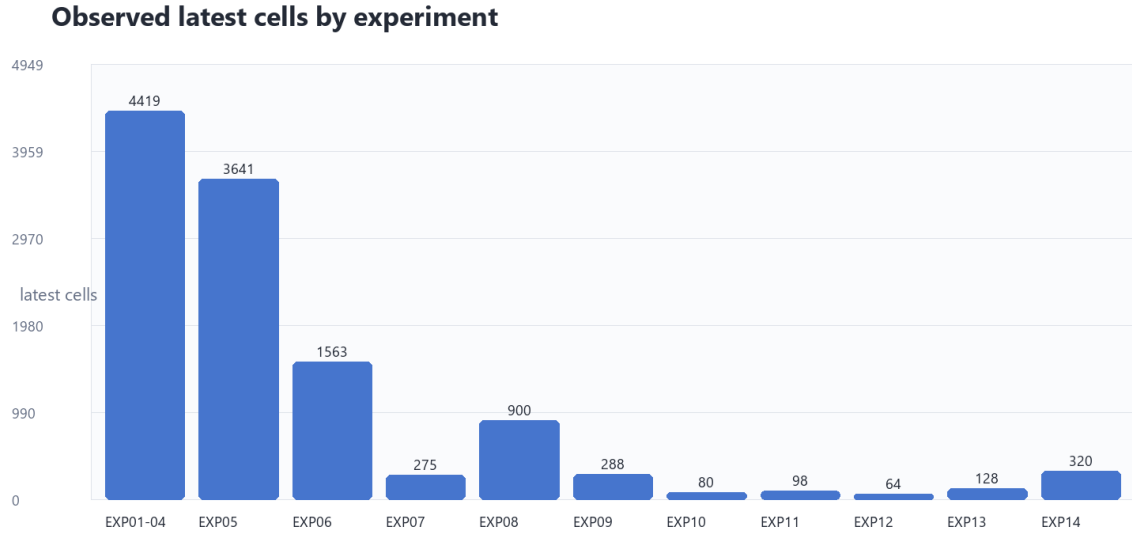


Figure 7: Observed latest cells by experiment. EXP01–EXP04 and EXP05 dominate raw scale; EXP09–EXP15 contribute stricter objective/tool validations.

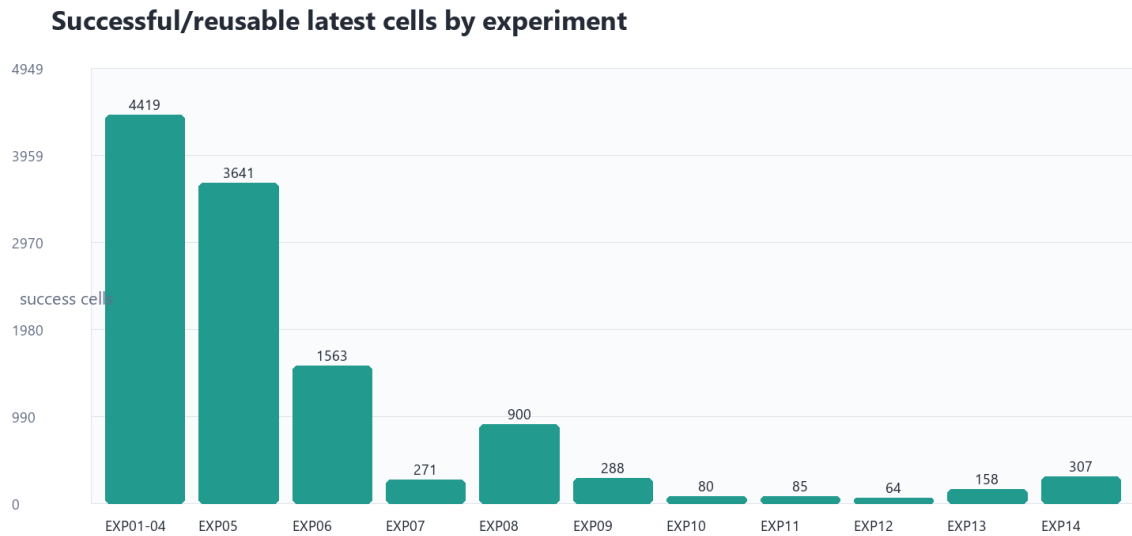


Figure 8: Successful or reusable latest cells by experiment. Later failures are more informative because validators become stricter.

12 Screenshots of public/page artifacts

The following screenshots are included to make clear that the later experiments produced concrete edited artifacts, not only abstract JSON evaluations.

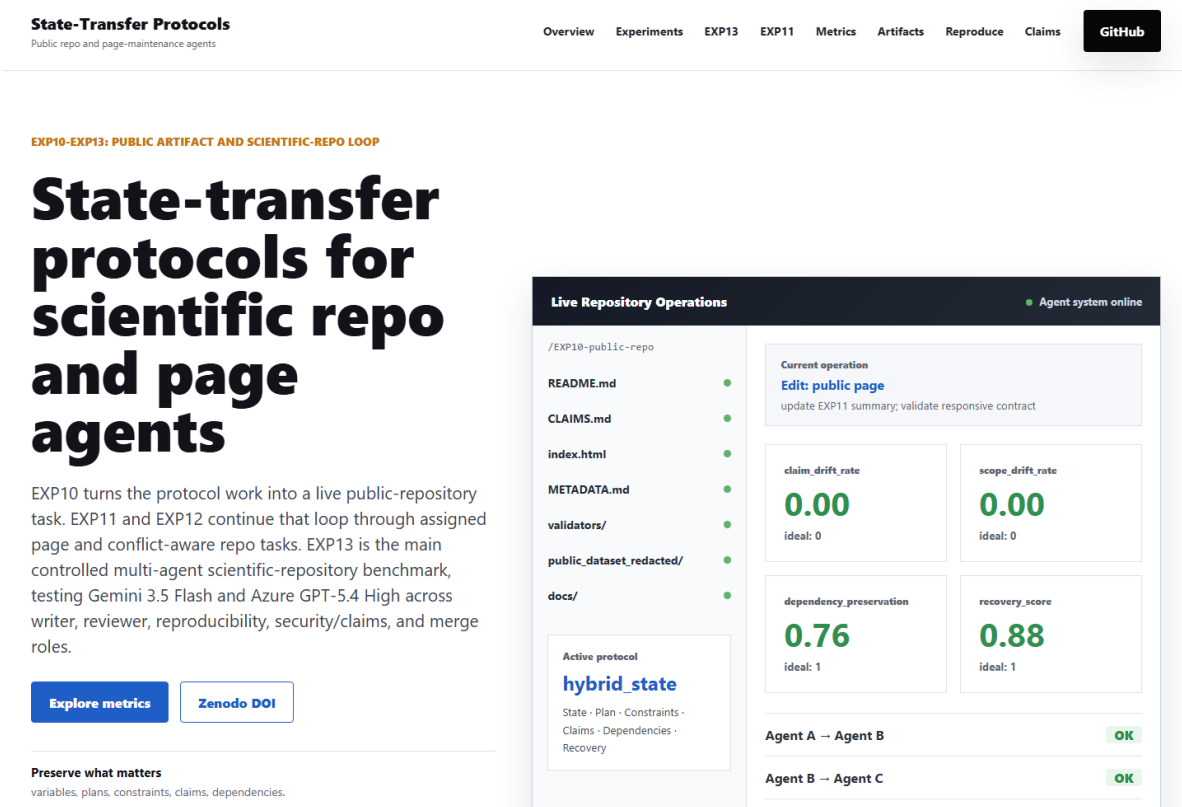


Figure 9: Public repository page. Captured from local HTML artifacts generated by the repo/tool-agent experiments.

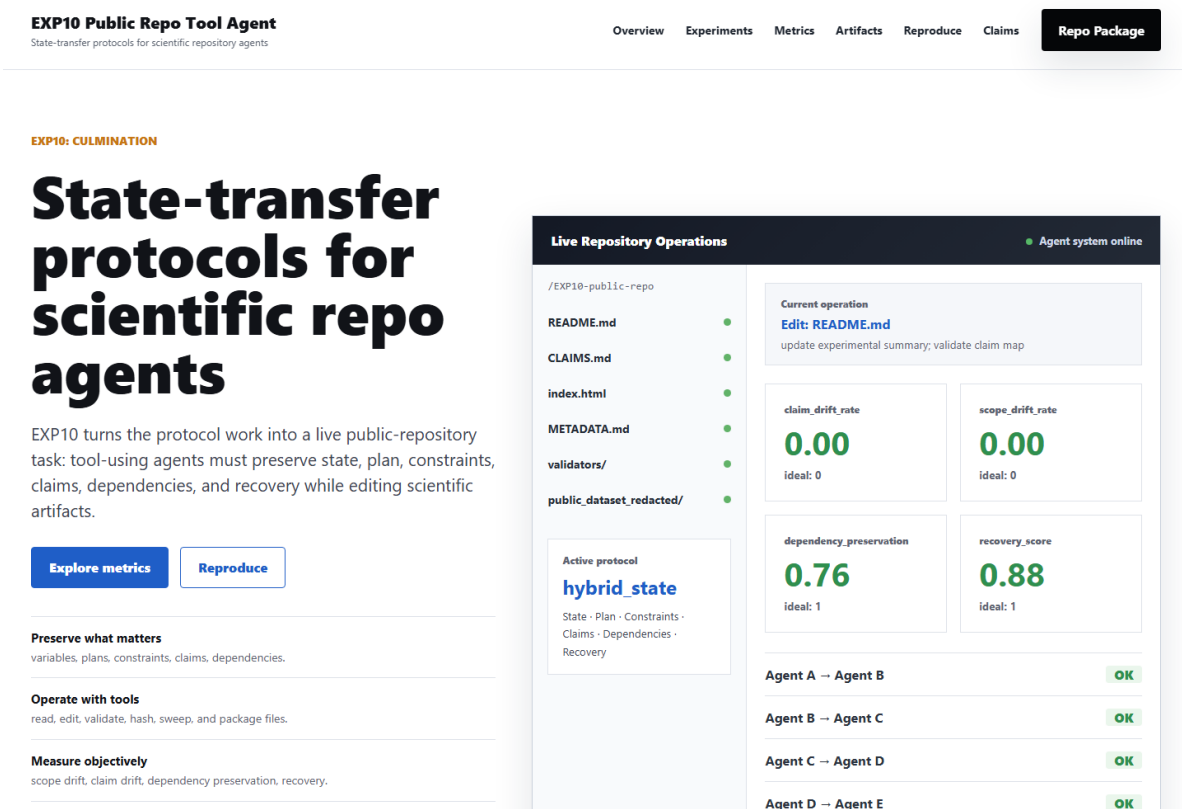


Figure 10: EXP10 public page artifact. Captured from local HTML artifacts generated by the repo/tool-agent experiments.

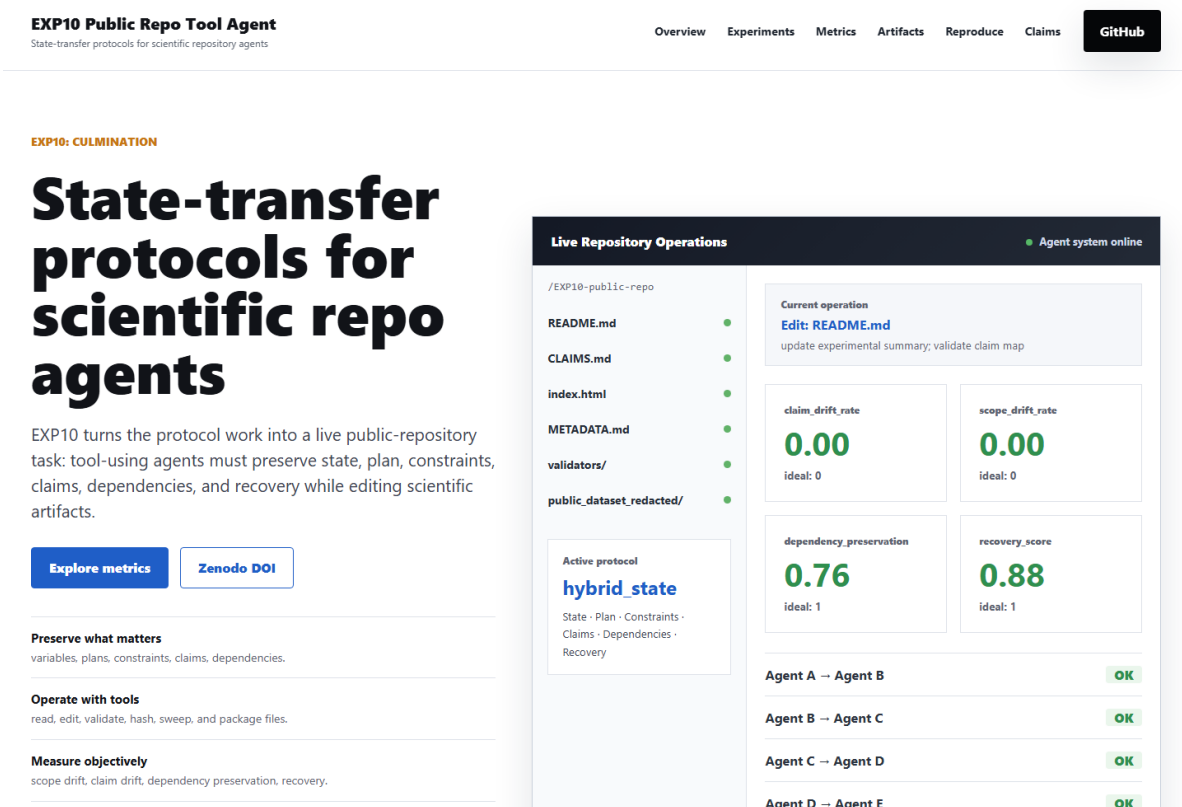


Figure 11: EXP13 scientific repo page. Captured from local HTML artifacts generated by the repo/tool-agent experiments.

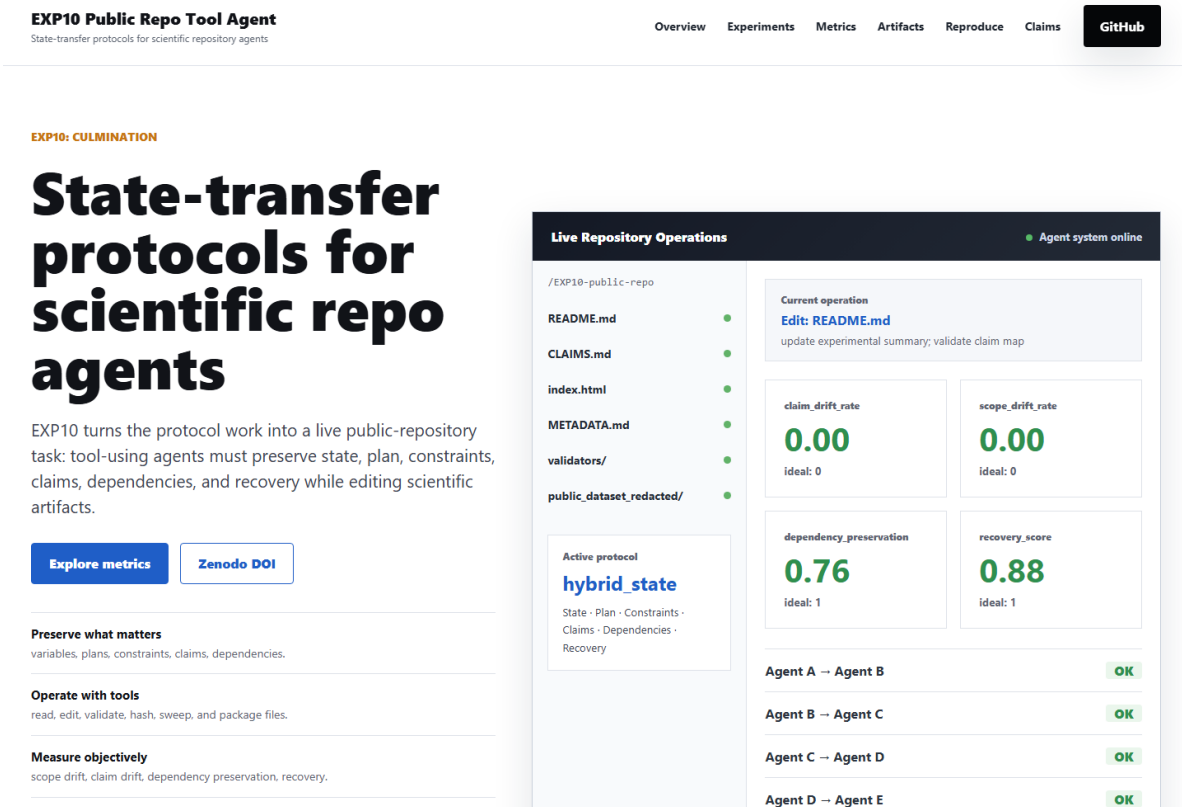


Figure 12: EXP14 long-horizon final page. Captured from local HTML artifacts generated by the repo/tool-agent experiments.

13 Synthesis and extrapolated interpretation

The strongest synthesis is a three-part story:

1. **compressed** is the universal operational baseline: portable, cheap, robust, and less fragile under strict contracts.
2. **hybrid.state** is not a global quality winner; it is a specialized state-transfer carrier. Its value appears when preserving constraints, current plan, dependencies, and handoff state matters more than prose quality.
3. Real agent systems turn prompt design into interface design. Failures increasingly come from schema mismatch, parser contract violations, blank outputs, repair loops, and scope drift rather than from ordinary semantic misunderstanding.

The extrapolated research direction is therefore not “prompt compression” alone. It is the design of compact operational protocols for agent continuity across models, languages, tools, repositories, and time.

14 Claim strength

- Strong: EXP05–EXP15 support **compressed** as a robust baseline.
- Strong but narrow: **hybrid.state** improves or protects state-oriented behavior in controlled/handoff settings, but carries tool-contract fragility.
- Strong: objective validators in EXP09–EXP15 reduce dependence on LLM-as-judge.
- Exploratory: ZH/tokenization claims remain promising but not closed without tokenizer-controlled analysis.
- Exploratory: visual quality comparisons require richer human or computer-vision scoring beyond current proxy metrics.

15 Limitations

This report aggregates heterogeneous experiments rather than running one monolithic preregistered benchmark. Some early results were recovered from historical logs; later experiments are cleaner and more validator-driven. Model routes are time-sensitive. Saturated success rates in EXP09, EXP10, and EXP12 should be interpreted as interface feasibility, not as evidence that the task class is solved. EXP14 and EXP15 indicate that long-horizon degradation remains open, and EXP15-B shows that model/interface compatibility can dominate visual/responsive task outcomes.

16 Reproducibility notes

Primary artifacts live under `Investigaciones/Protolenguaje_AgentesIA/05.Experimentos`. Public-facing release material lives under `13.Repo_Publico/GITHUB_RELEASE.state-transfer-protocols`. Frozen summaries, clean latest-cell files, task banks, validators, manifests, checksums, and reports exist for the major later experiments. This LaTeX report was generated from local files by `build_exp01_exp15_update.py` after the EXP15/EXP15-B final freezes.